

IPW3 : NACA0012 Code-to-Code Comparisons

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**POLYTECHNIQUE
MONTREAL**

TECHNOLOGICAL
UNIVERSITY



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IPW3 – Participants

ID	Participant / Organization
001	CIRA
002	AeroTex
003	NRC
004	ONERA
006	Dassault
007	Polytechnique Montréal
008	Collins Aerospace
009	Sikorsky
010	NASA
013	Boeing
014	Airbus
015	NIAR
019	Synopsys
020	Bombardier



Geometry and Aerodynamic Conditions



NACA0012 – Wind-Tunnel View

Geometry

Parameter	Value
Airfoil	NACA0012
Chord, c [m]	0.5334
Angle of attack, α [°]	4.0

Aerodynamic Conditions

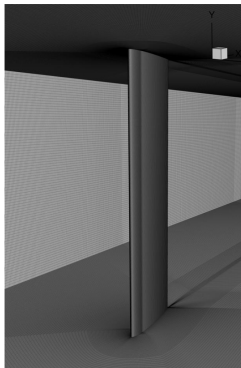
Parameter	Value
M_∞ [-]	0.3114
U_∞ [m/s]	102.15
T_∞ [K]	267.75
ρ_∞ [kg/m ³]	1.1884
Re_c [-]	3.8441×10^6

Quantities of Interest

$$C_L, \quad C_D, \quad C_M, \quad C_p$$



Icing Conditions



NACA0012 – Mesh Tunnel View

Grid	Nodes	Elements
L1	60.45 M	59.86 M
L2	27.31 M	26.99 M
L3	11.23 M	11.08 M
L4	5.95 M	5.86 M

	AE3932	AE3933
T_0 [°C]	−5.6	−2.2
T_∞ [K]	262.305	265.8
M_∞	0.3151	0.3127
U_∞ [m/s]	102.31	102.19
LWC [g/m ³]	0.55	0.55
MVD [μm]	20	20
Exposure time [min]	7	7
Exp. ice mass [g]	101.0	85.9

Experimental Observation

The warmer AE3933 condition produces approximately **15% less ice mass** than AE3932 (85.9 g vs. 101.0 g).

Droplet Distribution

- 1-, 3-, 7-, and 15-bin distributions
- Total LWC preserved across distributions
- Evaluated across all four grid levels

Quantities of Interest

$$\beta, \quad HTC, \quad T_s, \quad FF, \quad m_{\text{water}}, \quad m_{\text{ice}}, \quad m_{\text{evap}}$$



NACA0012 – Modeling Choices Across Submissions

- **Turbulence:** $k-\omega$ formulations were most common (**62.5%**), followed by SA (**37.5%**).
- **Droplet formulation:** Predominantly Eulerian (**75%**), with **25%** Lagrangian.
- **Heat transfer:** Documented approaches included recovery-temperature and boundary-layer integral methods.
- **Thermodynamics:** Messinger-based models were dominant (**71.4%**); SWIM-based and fluid-film approaches were also used.
- **Surface roughness:** Documented submissions used constant roughness heights ($k_s \approx 0.5$ mm).
- **Geometry evolution:** Most commonly based on Lagrangian/IB/node displacement; level-set, ALE morphing, and interpolation-based approaches were also reported.
- **Ice accretion strategy:** Both single-shot (**5 submissions**) and multilayer/multishot (**4 submissions**) approaches were represented.

Modeling choices disclosed by participants: 001, 002, 004, 007, 008, 009, 014, and 019.



NACA0012 – Summary of Submitted Data

AE3932

ID	L1				L2				L3				L4			
	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓				✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								✓
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008		*				*				*				*		
009		*				*				*				*		
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020				✓				✓	✓	✓	✓	✓				✓
Total	6	5	5	9	7	6	6	10	7	6	6	8	6	5	5	8

AE3933

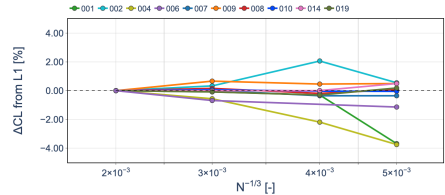
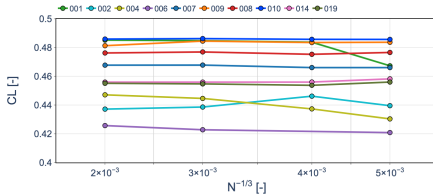
ID	L1				L2				L3				L4			
	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓	✓	✓	✓	✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								✓
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
009	✓				✓				✓				✓			
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020												✓				
Total	8	6	6	8	8	6	6	8	8	6	6	7	7	5	5	6

✓: populated CutData received for the corresponding droplet-bin distribution.

*: Excel results only, no CutData slices.



NACA0012 – Lift Coefficient Grid Convergence

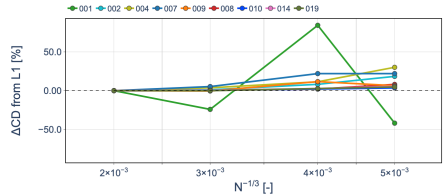
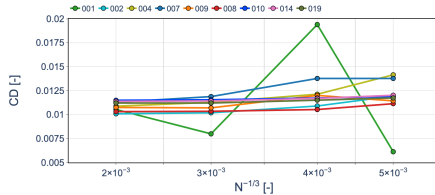


Observations

- At L1, the median lift coefficient is $C_L = 0.4677$, with an IQR of 0.0262 (5.6%).
- Most submissions exhibit weak sensitivity to spatial grid refinement.
- Relative to L1, coarse-grid differences remain within approximately 4% across the submissions.



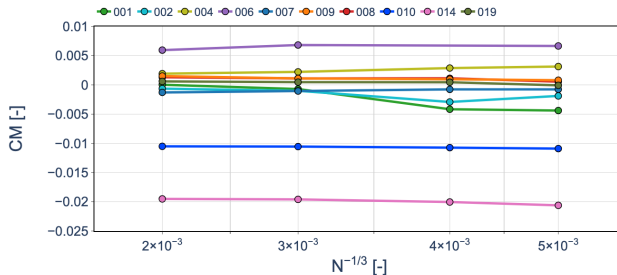
NACA0012 – Drag Coefficient Grid Convergence



Observations

- At L1, the median drag coefficient is $C_D = 0.01088$ (**108.8 drag counts**), with an IQR of **7.57 drag counts** (7.0%).
- C_D exhibits the greatest sensitivity to spatial grid refinement among the integrated aerodynamic coefficients.
- On the coarser grids, most submissions deviate by up to approximately **30%** from their respective L1 values.
- One submission additionally exhibits pronounced non-monotonic behavior with grid refinement.

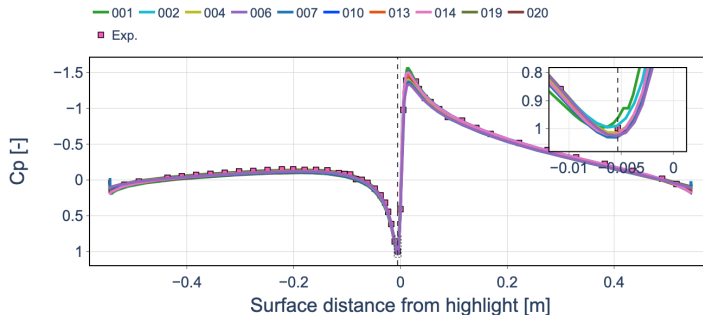
NACA0012 – Pitching Moment Grid Convergence



Observations

- At L1, the median pitching-moment coefficient is $C_M \approx 0$, with an IQR of 2.60×10^{-3} .
- Most submissions exhibit little variation in C_M across the grid hierarchy.

NACA0012 – Surface Pressure Distribution (L1)



Observations

- The L1 C_p distributions are tightly clustered and show good overall agreement with the experimental measurements.
- The largest code-to-code differences are localized near the leading-edge suction peak and stagnation region.
- Away from the leading edge, the predicted pressure distributions are nearly coincident across submissions.

NACA0012 – Aerodynamic Assessment Summary

L1 Code-to-Code Variability

	Median	IQR	Relative IQR
C_L	0.4677	0.0262	5.6%
C_D	108.8 counts	7.57 counts	7.0%
C_M	≈ 0	2.60×10^{-3}	–

Relative IQR is not reported for C_M because its median is close to zero.

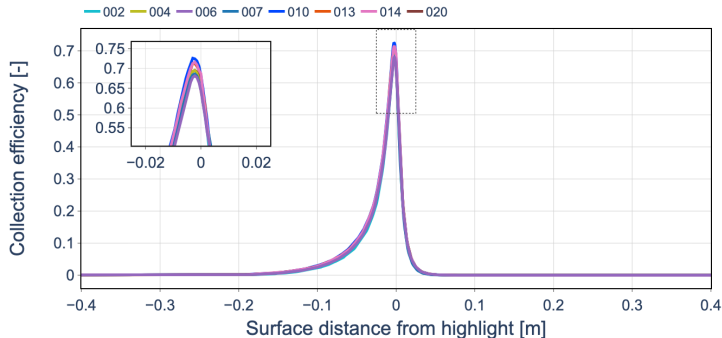
Grid-Refinement Behavior

- C_L and C_M exhibit limited within-code sensitivity to grid refinement for most submissions.
- C_D shows greater and less uniform grid sensitivity, including non-monotonic behavior for selected submissions.
- For most submissions, code-to-code differences exceed the changes produced by spatial refinement.

Aerodynamic Takeaway

Overall, the aerodynamic predictions are well clustered, with L1 C_p distributions showing good agreement with experiment despite differences in prescribed surface roughness. Among the integrated coefficients, C_D exhibits the greatest sensitivity to spatial grid refinement.

NACA0012 – Collection Efficiency (L1, 15 Bins)

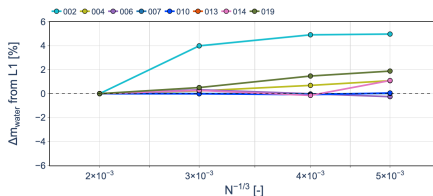
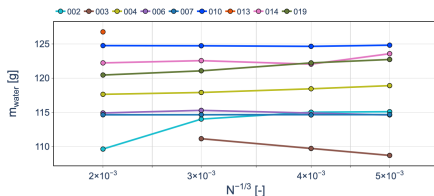


Observations

- The L1 collection-efficiency profiles are tightly clustered across submissions and exhibit very similar leading-edge behavior.
- Differences are concentrated near the peak region and impingement limits.



NACA0012 – Water Mass Grid Convergence (15 Bins)

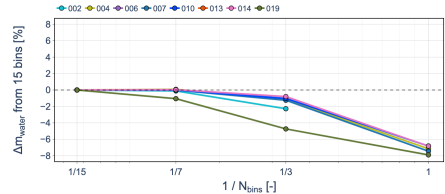
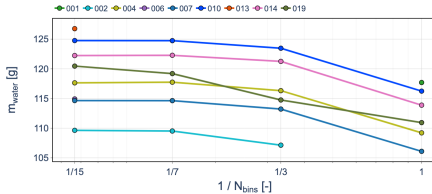


Observations

- For the 15-bin distribution, the L1 median impinging water mass is 119.05 g, with an IQR of 6.39 g (5.37%).
- Most submissions exhibit weak grid sensitivity and remain close to their L1 prediction across the grid hierarchy.



NACA0012 – Droplet Distribution Sensitivity (L1)

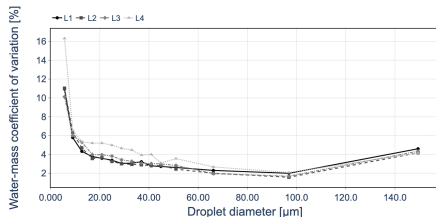


Observations

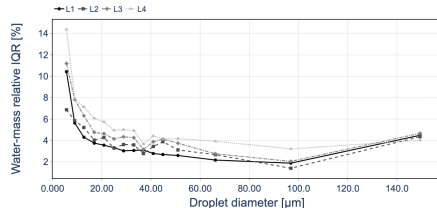
- The 15-bin and 7-bin distributions produce similar total impinged water masses.
- Reducing the distribution from 7 to 3 bins introduces only a modest additional change for most submissions.
- Relative to the 15-bin distribution, the single-bin approximation reduces the collected water mass by approximately 5–8% for most participants.



NACA0012 – Water-Mass Variability by Droplet Diameter (15 Bins)



Coefficient of Variation

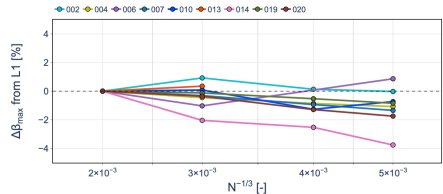
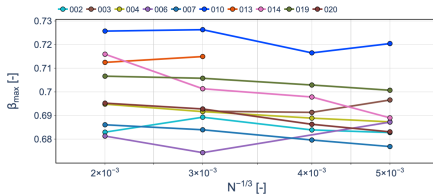


Relative IQR

Observations

- Code-to-code variability is strongly dependent on droplet diameter, with the largest spread observed for the smallest droplets
- Variability decreases rapidly with increasing diameter and remains comparatively low over most of the intermediate droplet-size range.
- The coarser L3–L4 grids generally exhibit greater variability than L1–L2, particularly for the smallest droplets.
- A slight increase in variability is observed again for the largest droplets.

NACA0012 – Peak Collection Efficiency Grid Convergence (15 Bins)

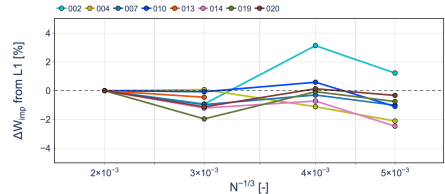
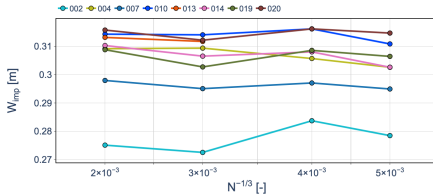


Observations

- At L1, the median peak collection efficiency is $\beta_{\max} = 0.695$, with an IQR of 0.0209 (3.0%).
- Most submissions exhibit limited sensitivity of $\beta_{\max}$ to spatial grid refinement.
- Similar code-to-code clustering and grid-refinement behavior are observed for the other droplet-size distributions (not shown).
- The location of $\beta_{\max}$ was also examined (not shown); however, it is strongly influenced by the local leading-edge surface discretization. Consequently, no clear grid-convergence conclusion is drawn for the peak position.



NACA0012 – Impingement Width Grid Convergence (15 Bins)



Observations

- At L1, the median impingement width is 0.309 m, with an IQR of 8.93×10^{-3} m (2.9%).
- The impingement width is not fully grid-converged, although the variations between successive grid levels remain relatively small for most submissions.
- The remaining grid dependence partly reflect the sensitivity of the impingement limits to the surface discretization.



NACA0012 – Droplet Impingement Assessment

Spatial Grid Convergence

- Water mass, $\beta_{\max}$, and impingement width show limited grid sensitivity for most submissions.
- Code-to-code variability is generally comparable to or larger than the grid-refinement effect.
- Peak position was examined, but its sensitivity to the local leading-edge discretization prevented a clear convergence assessment.

Droplet-Size Effects

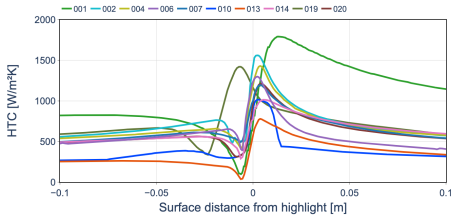
- The 7-bin distribution closely reproduces the 15-bin results, while the single-bin approximation systematically underpredicts collected water mass.
- Code-to-code variability is largest for the smallest droplets and decreases substantially over the intermediate diameter range.

Impingement Takeaway

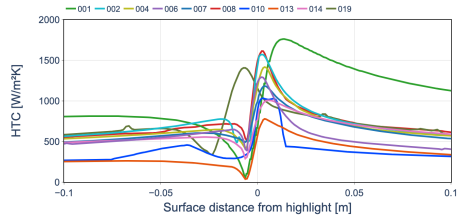
Overall, the impingement predictions are relatively consistent with spatial grid refinement. Droplet-size discretization has a clearer systematic influence, with 7 bins providing results close to the 15-bin reference.



NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932



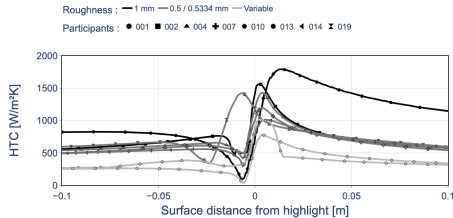
AE3933

Observations

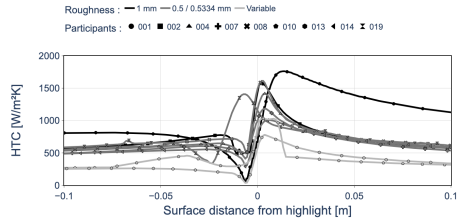
- Considerable code-to-code variability is observed in *HTC*, particularly near the leading edge, with similar behavior for both icing conditions.
- Prescribed surface roughness contributes systematically to the spread, while the remaining code-to-code variability must also be influenced by differences in the *HTC* computation methodology.



NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932 – Grouped by Roughness



AE3933 – Grouped by Roughness

Observations

- Considerable code-to-code variability is observed in HTC , particularly near the leading edge, with similar behavior for both icing conditions.
- Prescribed surface roughness contributes systematically to the spread, while the remaining code-to-code variability must also be influenced by differences in the HTC computation methodology.



NACA0012 – Integrated Convective Heat Transfer

Grid-Convergence Metric

To assess whether the differences in the local HTC distributions translate into a clear grid-convergence behavior, an integrated convective heat-transfer quantity is evaluated along the surface slice:

$$Q'_c = \int HTC(s) [T_s(s) - T_{rec}(s)] ds \quad [Q'_c] = W/m.$$

The submitted HTC and surface temperature T_s are used directly. The recovery temperature T_{rec} , however, was not part of the submitted dataset and must therefore be reconstructed.

Recovery Temperature Reconstruction

The submitted pressure coefficient is used to estimate the local external-flow state and subsequently the recovery temperature:

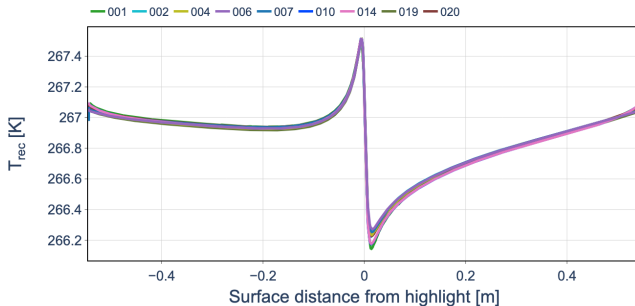
$$C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{rec}(s)$$

Assuming $p_s \simeq p_e$ and an isentropic external flow,

$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_{p,e}, \quad M_e^2 = \frac{2}{\gamma - 1} \left[\left(\frac{p_0}{p_e} \right)^{(\gamma-1)/\gamma} - 1 \right],$$

$$T_e = \frac{T_0}{1 + \frac{\gamma-1}{2} M_e^2}, \quad T_{rec} = T_e \left[1 + r \frac{\gamma-1}{2} M_e^2 \right], \quad r = Pr^{1/3} \approx 0.9.$$

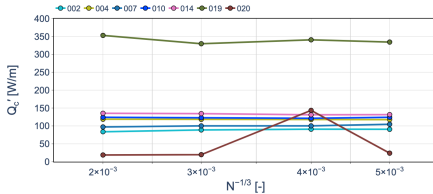
NACA0012 – Reconstructed Recovery Temperature (L1)



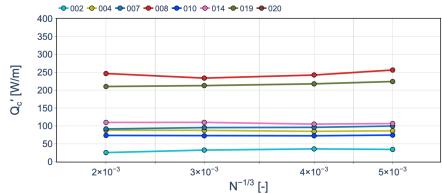
Observations

- The reconstructed T_{rec} distributions are closely clustered across submissions.
- This consistency follows directly from the close agreement previously observed in the submitted C_p distributions.
- Differences in Q'_c are therefore expected to originate primarily from HTC and T_s , rather than from the reconstructed T_{rec} .

NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932



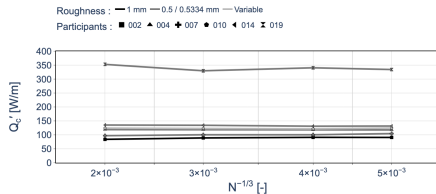
AE3933

Observations

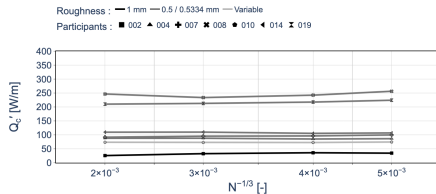
- Most submissions exhibit limited grid sensitivity in the integrated Q'_c .
- Grouping by prescribed roughness reveals a systematic roughness effect, although significant code-to-code variability remains within the roughness groups.



NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932 – Grouped by Roughness



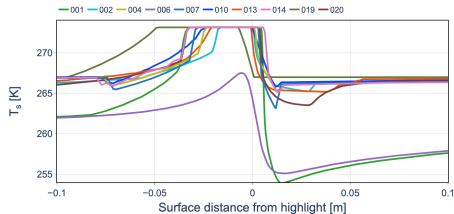
AE3933 – Grouped by Roughness

Observations

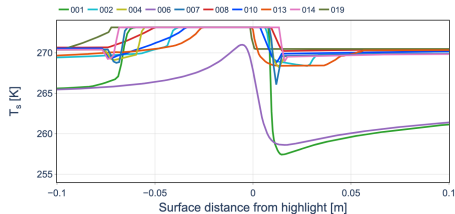
- Most submissions exhibit limited grid sensitivity in the integrated Q'_c .
- Grouping by prescribed roughness reveals a systematic roughness effect, although significant code-to-code variability remains within the roughness groups.



NACA0012 – Surface Temperature Comparison (L1)



AE3932

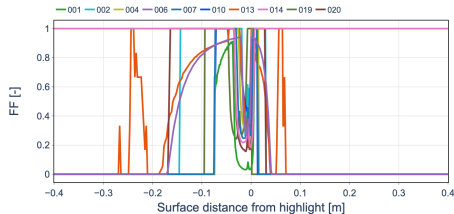


AE3933

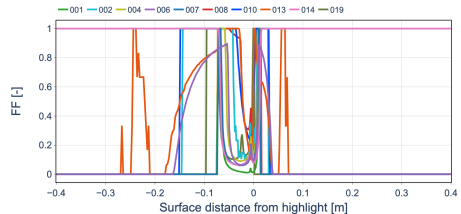
Observations

- Substantial code-to-code variability is observed in the surface-temperature distributions for both conditions.
- AE3933 exhibits a wider region where the surface temperature remains near the freezing temperature, consistent with its warmer freestream condition.
- The extent of this near-freezing region directly influences the predicted freezing-fraction distribution and, ultimately, the ice-accretion limits.

NACA0012 – Freezing Fraction Comparison (L1)



AE3932



AE3933

Observations

- Considerable code-to-code variability is observed in the freezing-fraction distributions for both conditions.
- AE3932 generally exhibits a narrower low-freezing-fraction region near the leading edge, whereas the warmer AE3933 condition tends to produce a wider region.
- This trend is consistent with the broader near-freezing surface-temperature region observed for AE3933.
- However, the large code-to-code variability and differences in transition locations make the freezing-fraction distributions difficult to interpret quantitatively.

NACA0012 – HTC & Thermodynamics Assessment

Convective Heat Transfer

- Local HTC exhibits substantial code-to-code variability, with prescribed roughness explaining only part of the spread.
- Reconstructed T_{rec} is closely clustered, consistent with the agreement previously observed in C_p .
- Despite the local HTC differences, integrated Q'_c exhibits limited grid sensitivity for most submissions.

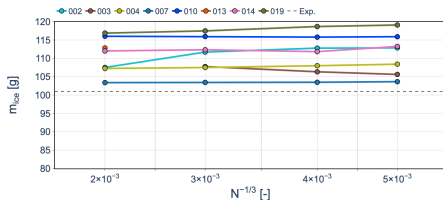
Thermodynamic Response

- Surface temperature and freezing fraction exhibit substantially greater code-to-code variability.
- The warmer AE3933 condition generally produces a wider near-freezing surface-temperature region and a correspondingly wider low-freezing-fraction region.
- The large variability in freezing fraction and transition locations limits more quantitative code-to-code interpretation.

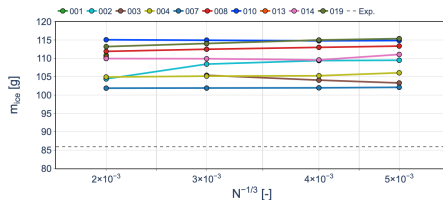
Takeaway

Convective heat transfer is relatively insensitive to grid refinement but exhibits substantial code-to-code variability. The warmer AE3933 condition produces a broader near-freezing region, which is expected to influence the freezing behavior and subsequent ice-accretion limits.

NACA0012 – Ice Mass Grid Convergence (15 Bins)



AE3932



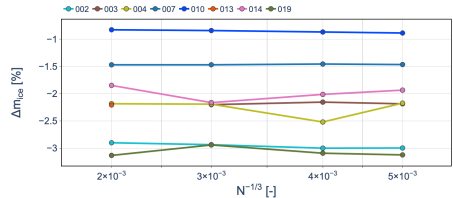
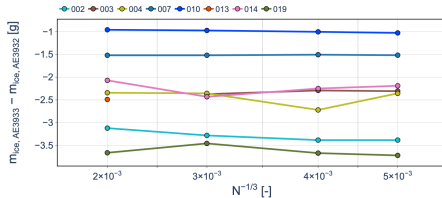
AE3933

Observations

- Most submissions exhibit limited sensitivity of the predicted ice mass to spatial grid refinement for the 15-bin distribution.
- AE3933 generally produces less ice mass than AE3932, with predicted reductions of at most approximately 3%, compared with the approximately 15% observed experimentally.



NACA0012 – Ice Mass Difference Between Conditions (15 Bins)

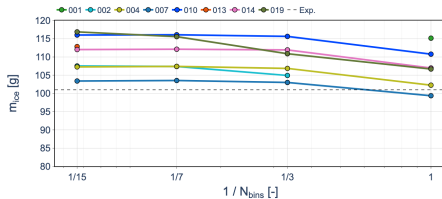


Observations

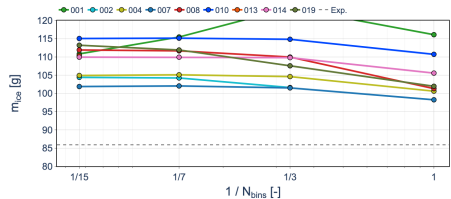
- The matched-participant comparison confirms systematically lower ice mass for AE3933 than AE3932 for most submissions.
- The numerical predictions reproduce the experimental trend of lower ice mass for AE3933, but predict a substantially smaller difference than the approximately 15% observed experimentally.



NACA0012 – Ice Mass Sensitivity to Droplet Distribution (L1)



AE3932



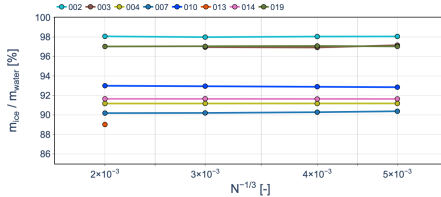
AE3933

Observations

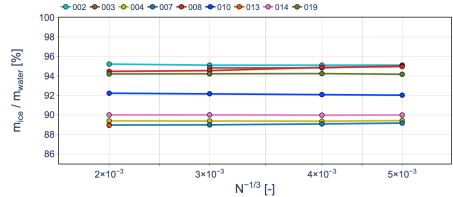
- The 15- and 7-bin distributions generally produce comparable ice masses for both conditions.
- Sensitivity increases when the distribution is reduced to 3 bins and becomes more pronounced for the single-bin representation.



NACA0012 – Ice-to-Water Mass Ratio



AE3932



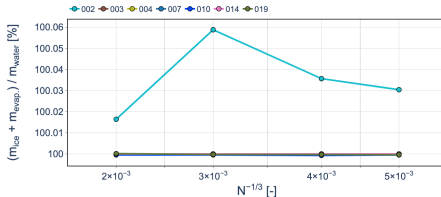
AE3933

Observations

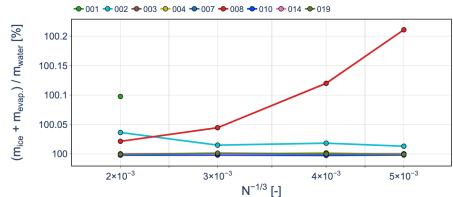
- Approximately 90% of the impinging water is converted into ice for both conditions.
- AE3933 consistently exhibits a lower ice-to-water ratio than AE3932, indicating greater water loss.
- Since the water impingement is nearly identical between the two conditions, this difference originates primarily from the thermodynamic mass balance.



NACA0012 – Ice and Evaporation-to-Water Mass Ratio



AE3932



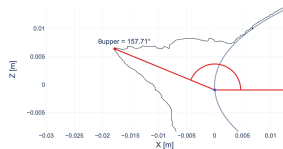
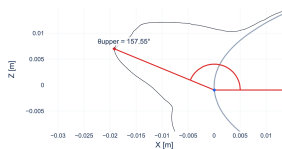
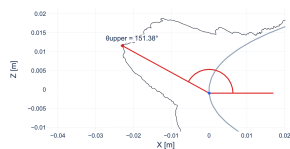
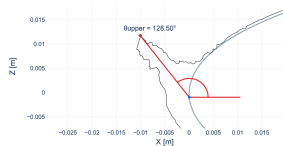
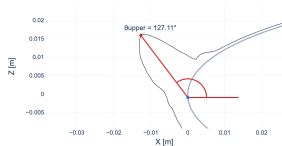
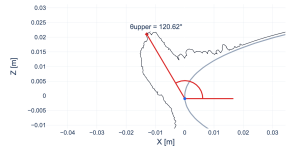
AE3933

Observations

- The combined ice and evaporated-water mass accounts for approximately 100% of the impinged water for most submissions.
- This indicates that the lower ice-to-water ratio for AE3933 is primarily associated with increased evaporation.
- Values slightly above 100% are observed for a few submissions and are likely attributable to numerical integration or round-off errors.



NACA0012 – Experimental Upper Horn Angle

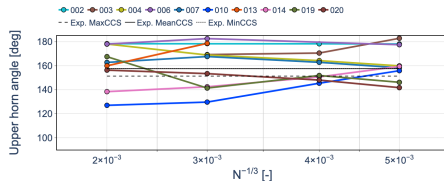
**MinCCS****MeanCCS
AE3932****MaxCCS****MinCCS****MeanCCS
AE3933****MaxCCS**

Experimental Reference

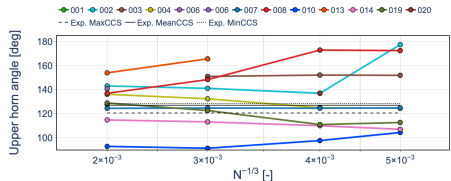
- The experimental upper-horn angle ranges from approximately $151\text{--}158^\circ$ for AE3932 and $121\text{--}129^\circ$ for AE3933.



NACA0012 – Upper Horn Angle Grid Convergence (15 Bins)



AE3932

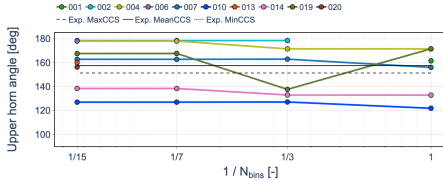


AE3933

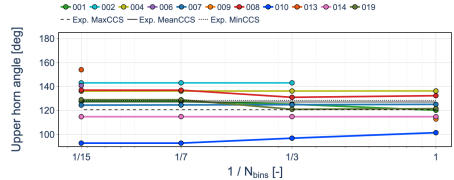
Observations

- Horn-angle predictions exhibit substantial code-to-code variability and do not show systematic grid convergence across the participant ensemble.
- AE3932 becomes more tightly clustered on the coarser grids, whereas AE3933 exhibits increasing spread with grid coarsening.
- A consistently lower upper-horn angle is observed for AE3933 compared with AE3932, in agreement with the experimental trend.
- At L1, the code-to-code scatter remains comparable between conditions, with relative IQRs of approximately 16% for AE3932 and 13% for AE3933.

NACA0012 – Upper Horn Angle Sensitivity to Droplet Distribution (L1)



AE3932



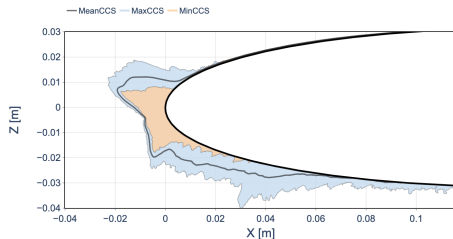
AE3933

Observations

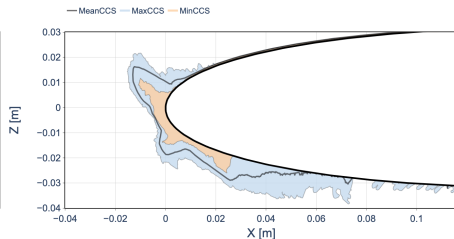
- The upper-horn angle exhibits only small variations across the different droplet-size distributions for most submissions.
- The overall horn orientation is therefore relatively insensitive to the droplet-size discretization.



NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932 – Experiment

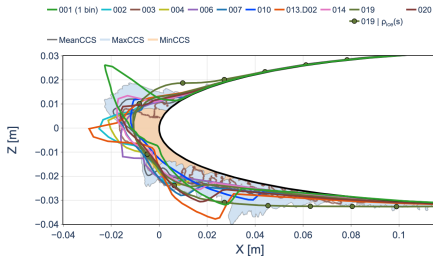


AE3933 – Experiment

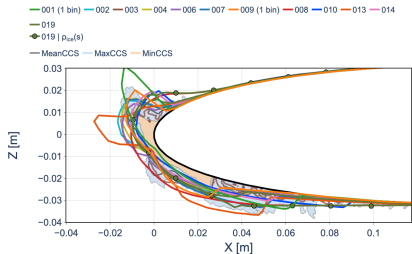
Observations

- The experimental ice shapes show a clear geometric distinction between AE3932 and AE3933, particularly in upper-horn orientation.
- The code-to-code variability observed in the preceding analyses is ultimately reflected in the variability of the predicted ice shapes and horn angles.
- The experimental condition-to-condition distinction is reproduced to varying degrees by the numerical predictions.
- Grouping by prescribed roughness reveals some systematic roughness dependence.

NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932

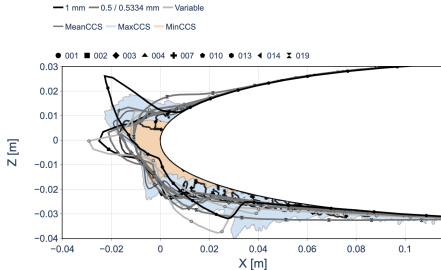


AE3933

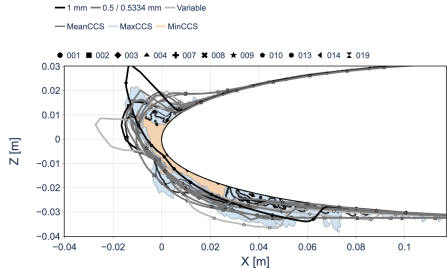
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NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932 – Grouped by Roughness

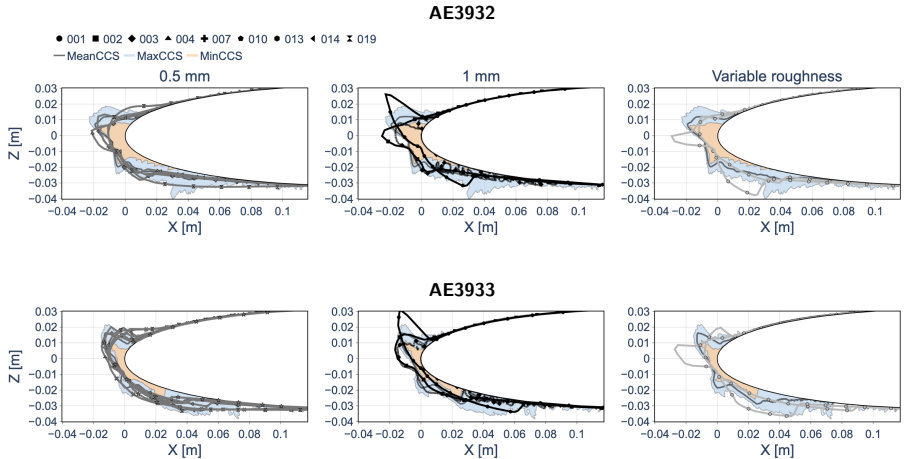


AE3933 – Grouped by Roughness

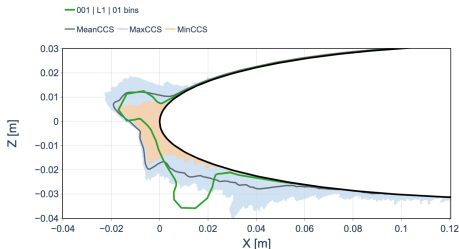
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- The experimental condition-to-condition distinction is reproduced to varying degrees by the numerical predictions.
- Grouping by prescribed roughness reveals some systematic roughness dependence.

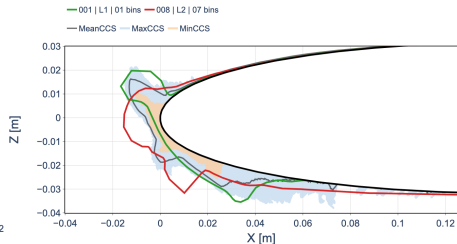
NACA0012 – Effect of Surface Roughness on Ice Shapes (L2, 15 Bins)



NACA0012 – Multilayer Ice-Shape Comparison



AE3932



AE3933

Observations

- The available multilayer predictions suggest improved agreement with the experimental ice shapes and capture more of the observed geometric features than the single-layer results.
- The limited number of multilayer submissions, however, prevents drawing a conclusive assessment of this improvement across the participant ensemble.



NACA0012 – Ice Accretion Summary

Verification

- Ice mass shows limited grid sensitivity, with an L1 code-to-code IQR of approximately 5% for both icing conditions; horn orientation exhibits greater variability, while the 15- and 7-bin distributions generally provide comparable results.

Condition-to-Condition Response

- AE3933 generally produces less ice than AE3932, although the predicted reduction remains substantially smaller than the approximately 15% observed experimentally.
- A lower upper-horn angle is consistently predicted for AE3933, reproducing the experimentally observed geometric shift between the two conditions.

Ice Geometry

- Considerable code-to-code variability remains in the predicted ice shapes, particularly in horn orientation and local geometry.
- The available multilayer predictions suggest improved agreement with experiment and capture additional geometric features, although the limited number of submissions prevents a conclusive assessment.

Takeaway

The main distinction between AE3932 and AE3933 emerges in the thermodynamic response and resulting ice accretion, rather than in the upstream aerodynamics or water impingement.

NACA0012 – Overall Assessment

Aerodynamics

- Overall, the aerodynamic predictions are well clustered; C_p agrees well with experiment despite differences in prescribed roughness, while C_D exhibits the greatest sensitivity to spatial refinement.

Droplet Impingement

- Predictions remain relatively consistent across the grid hierarchy, with small grid-to-grid variations, and 15- and 7-bin distributions provide comparable results.

Convective Heat Transfer & Thermodynamics

- HTC exhibits substantial code-to-code variability, while the warmer AE3933 condition produces the expected broader near-freezing surface-temperature region.

Ice Accretion

- AE3933 consistently produces less ice than AE3932, although the predicted reduction remains well below the approximately 15% observed experimentally.
- A lower upper-horn angle is consistently predicted for AE3933, reproducing the experimentally observed geometric shift between the two conditions.
- Considerable code-to-code variability remains in the detailed ice geometry, while the available multilayer predictions suggest improved capture of experimental geometric features.